

Related Equipment/Instruments/Materials/Miscellaneous:

Equipment: Acrylic curing unit, Casting machine, De-waxing and porcelain furnace, Light curing unit, Light and electron microscope, Electro-polishing unit with Acid solution, Sand Blaster, CAD-CAM machine, EMG-JVA (Bio-tens), Model trimmer, ceramage machine.

Instrument: Mirror, Probe, Twiser, Excavator, Perio- Probe, Mould (Partial, Complete, Dentate, Typodont), Slide caliper, Willis gauge, metal-gauge, Fox's plane, Articulators (Non-Anatomical, Semi-adjustable, Anatomical articulator with face-bow transfer, accessories, Half-articulator, Impression tray (Edentulous tray, Stock- tray-Edentulous, Dentulous), Micro-motor, Portable turbine set with Hand pieces, Agar-bath, Duplicating flask, Wax knife, curver, Electric wax knife, Model- saw, Dental Press, Dental Flask, Surgical kit for implant placement, Rubber bowl, Spatula, Different Dental surveyor, Gothic arch, Different Polishing kits.

Materials: Impression (Compound- sheet and stick variety, Alginate, Rubber base, Agar-agar, Zinc oxide eugenole), Gypsum products (Ordinary plaster, Hard plaster, Investment Plaster, Impression plaster), Dental waxes (Modeling wax, Different Inlay waxes, Different Sheet casting waxes, Wax-mesh, Impression wax, Impression – Gutta- percha), Metal and Dental alloys- gold alloys, Acrylic resin (Self cure acrylic resin, Heat cure acrylic resin, Light activated AR), Flexible nylon denture base material, Cobalt-chromium alloy, Dental Implants, Die-pin, Gingival retraction cord, Dental Porcelain, Different SS wire, Vaseline, Shade guide (Spectrophotometry, Die spacer, Debubliizer, Porcelain Brushes, Laboratory composite and its accessories, Luting agents, Materials used in maxillofacial prosthesis (Acrylic resin, Elastomer), Soft lining materials, Acrylic and Porcelain and teeth, Precision attachments, Polishing materials for resin, metal and alloys, different separating media., ceramage powder and liquid.

Miscellaneous: Hand gloves, Face-mask, Hand wash, Soaps, Cotton, Suction tips, Napkins, Chest piece, Gluter-aldehyde solution, Patient apron

Professional Examination:**Marks distribution of Assessment of Prosthodontics****Total marks : 300**

- Written =100 (70 on SAQ +20 on MCQ + 10 on Formative)
- Oral (SOE) = 100
- Practical =100 (10 spotting, 20 short case, 30 long case, 30 lab work, 10 assignment)

Learning Objectives and Course Contents in Prosthodontics

To be taught in 4th phase

Pre-Clinical Prosthodontics

Learning Objectives	Contents	Teaching Hours
Student should be able to- Describe applied anatomy and physiology related to prosthodontics.	Applied anatomy and physiology related to prosthodontics	2 hrs lecture 1 hr tutorial 1 hr practical
Mention the materials and instruments used in Prosthodontics.	Materials and instruments used in prosthodontics	3 hrs lecture 2 hrs demonstration
- Identify the partially edentulous model - Perform impression, surveying and designing of partial denture framework. - Mention and perform laboratory steps for cast partial denture on partially edentulous cast.	Fabrication of Removable Partial Dentures using partial edentulous model: Impression (Demonstration only), surveying, designing partial denture framework, laboratory steps for cast partial denture on partially edentulous cast	3 hrs lecture 1 hr tutorial 1 hrs practical 1hrs demonstration
- Identify the complete edentulous model. - Perform preliminary impression taking, cast preparation, special tray preparation, final impression. - Prepare master cast preparation by boxing technique, temporary base preparation, occlusal rim preparation, jaw relation recording, transfer jaw relation record on articulator. - Select and alline teeth, arrangement of teeth, waxing, carving and laboratory procedures.	- Fabrication of complete denture prosthesis using edentulous models (cast): - Preliminary impression (Demonstration only)), cast preparation, special tray preparation, final impression, master cast preparation by boxing technique (only Demonstration), temporary base preparation, occlusal rim preparation, jaw relation recording (Demonstration), transfer jaw relation record on articulator, selection of teeth, arrangement of teeth, waxing, carving and laboratory procedures.	4 hrs lecture 4 hrs tutorial 2 hrs practical 2 hrs demonstration

Learning Objectives	Contents	Teaching Hours
Student should be able to- - Identify the model for fixed dental prosthesis, - Prepare the tooth on model. - Prepare impression taking from model. - Construct Model and die for laboratory technique.	Fixed Prosthodontics: Tooth preparation on model for crown or fixed partial denture, Impression taking (Demonstration only), model and die preparation, laboratory procedures	2 hrs lecture 1 hrs practical 1 hrs demonstration
- Identify the model for maxillofacial prosthesis, - Perform the impression taking, model preparation and fabrication of prosthesis.	Maxillofacial prosthesis in model: Impression taking, model preparation and fabrication of prosthesis	2 hrs lecture 1 hr tutorial 1 hrs practical
- Identify the model for implant - Describe implant procedure and prosthesis preparation.	Implant Dentistry: Implant procedure in model and prosthesis preparation	2 hrs lecture 1 hr tutorial 1 hrs practical

Learning Objectives and Course Contents in Prosthodontics

To be taught in 4th phase

Removable Partial Denture Prosthodontics

Learning Objectives	Contents	Teaching Hours
Student should be able to- - Enumerate the definition & terminology of RPD. - Examine and diagnose the cases for RPD.	Introduction, examination and diagnosis	2 hrs. lecture 1 hr tutorial 1 hr practical
- Formulate a treatment plan. - Perform the mouth preparation to improve the foundation of RPD.	Treatment planning and mouth preparation	3 hrs lecture 1 hrs practical
- Identify the landmark of imprints of upper and lower Jaw. - Perform model preparation	Impression and model	2 hrs lecture 1 hr tutorial 1 hr practical
- Describe the parts of surveyor. - Perform the technique of surveying. - Prepare a master cast.	Model surveying and preparation of master cast	3 hrs lecture 1 hrs tutorial 1 hrs practical
- Describe the classification of RPD. - Enumerate component parts of RPD.	Classification and component parts of RPD	2 hrs lecture 1 hr tutorial 1 hr practical
Outline the denture.	Designing partial denture	1 hr lecture 1 hr tutorial 1 hrs practical
Construct of working model.	Duplication	1 hr lecture 1 hrs practical
Perform wax pattern.	Wax pattern for cast partial denture	2 hrs lecture 1 hr tutorial 1 hr practical
- Performspruing, investing, burnout and casting - Perform finishing and polishing of partial denture frame work.	Casting procedure	2 hrs lecture 1 hr tutorial 1 hrs practical

Learning Objectives	Contents	Teaching Hours
Student should be able to- - Enumerate the occlusal rim - Perform occlusal rims on record bases for jaw relation	Occlusal rim	1 hr lecture 1 hr tutorial 1 hrs practical
- Explain the aesthetic and functional aspect of arranging teeth. - Perform alignment of teeth	Arranging teeth	1 hr lecture 1 hr tutorial 2 hrs practical
To check the denture in respect of functional and aesthetic consideration.	Trial	1 hr lecture 1 hr tutorial 1hr practical
Perform acrylic curing procedure.	Processing	1 hr lecture 1 hr tutorial 1 hrs practical
- Enumerateselective grinding. - Perform selective grinding.	Remounting	1 hr lecture 1 hrs practical
- Perform finishing, polishing - Determine the presence of any blebs arising from air inclusion.	Finishing and polishing	1 hr lecture 1 hrs practical
Instruct the patient about insertion, removal and maintenance	Insertion into the mouth	2 hrs lecture 1 hr tutorial 1 hrs practical
Identify the discrepancies for correction.	Review stage of treatment	1 hr lecture 1 hr tutorial 1 hr practical
Restore immediate aesthetic and functional rehabilitation.	Immediate removable partial denture	2 hrs lecture 1 hr tutorial 1 hrs practical

Learning Objectives and Course Contents in Prosthodontics

Complete denture Prosthodontics to be taught in 4th phase

Learning Objectives	Contents	Teaching Hours
Student should be able to- - Define complete denture prosthesis. - Enumerate the parts and surfaces of complete denture prosthesis. - Describe terminology.	Introduction of complete denture prosthesis	3 hrs lecture 1 hr tutorial 1 hrs practical
- Enumerate the different anatomical landmarks - Describe the denture bearing areas.	Anatomy and physiology of the tissues in the denture bearing areas and related structures	3 hrs lecture 1 hrs tutorial
- Define retention, - Describe the retaining and dislodging forces.	Principles of retention	3 hrs lecture 2 hrs tutorial
- Perform the history taking and clinical examination - Identify the potential problem areas - Improve the treatment.	History, clinical examination & diagnosis	4 hrs lecture 1 hr tutorial 4 hrs clinical
- Define impression - Perform the taking of impression. - Identify the common faults in the impression. - Correct the problem.	Primary Impression procedure	4 hrs lecture 1 hr tutorial 6 hrs clinical 1 hrs practical
- Define mouth preparation - Describe correction of tissue discrepancies. - Perform the surgical & non surgical correction	Mouth preparation	2 hrs lecture 3 hrs clinical
- Define the special tray. - Describe the materials use for special tray preparation. - Perform the fabrication of special tray	Special tray preparation	2 hrs lecture 1 hrs practical

Learning Objectives	Contents	Teaching Hours
Student should be able to- - Describe the materials use for final impression technique & master cast - Describe the procedure of final impression&master cast - Perform final impression&mastercast making.	Taking final impression and master cast preparation	2 hrs lecture 1 hrs practical
- Describe the denture base material & the materials use for construction of occlusal rim. - Perform the fabrication of denture base &occlusal rim.	Fabrication of denture base and occlusal rim	4 hours lecture 2 hrs practical
- Describe the jaw registration - Determining the procedure of vertical height. - Describe the effects of discrepancies of vertical height.	Registration of the jaw relationship	5 hrs lecture 1 hr tutorial 4 hrs clinical 1 hrs practical
- Define the concept of occlusions. - Describe the occlusion. - Perform the mounting	Mounting on articulator and concepts of occlusion	5 hrs lecture 1 hrs tutorial 1 hrs practical
- Select the teeth for individual mouth. - List out the point during selection of teeth. - Describe the main guidelines during alignment of teeth.	Selection and alignment of teeth	4 hrs lecture 2 hrs practical
- Determine the upper and lower denture for retention, support, stability - Check the occlusion - Correct it there is any discrepancies.	Trial of complete denture	4 hrs lecture 1 hrs tutorial 3 hrs clinical 1 hr practical

Learning Objectives	Contents	Teaching Hours
Student should be able to- - Define remounting - Perform remounting on articulator	Remounting on articulator.	3 hrs lecture 1 hr practical
Instruct the patient about the use and care of the denture.	Insertion of finished denture and follow up	3 hrs lecture 2 hrs clinical
- Describe the denture induced problems - Manage denture induce problems	Complaints of the complete denture	4 hrs lecture 2 hrs clinical 1 hr practical
- Define relining, rebasing and repairing. - Describe the technique of relining, rebasing and repairing. - Perform relining, rebasing and repairing.	Relining, rebasing and repairing	4 hrs lecture 1 hrs tutorial 2 hr clinical 2 hrs practical
- Define immediate denture - Identify the cases for immediate denture - Perform the immediate denture.	Immediate denture	4 hrs lecture 2 hr tutorial 2 hrs clinical 1 hr practical
- Define over denture - Identify the cases for over denture - Diagnose the case, designing of denture, preparation of abutment and fabrication.	Over denture	4 hrs lecture 2 hr tutorial 1 hr practical 2 hrs clinical

Learning Objectives and Course Contents in Prosthodontics (Fixed Prosthodontics) To be taught in 4th Phase

Learning Objectives	Contents	Teaching Hours
Student should be able to- - Define crown, inlay, onlay. - Describe the indication, contraindication, advantages, disadvantages of full/partial veneer crown.	- Intra-coronal and extra-coronal restoration: - Introduction, terminology, indication, contraindication, advantages & disadvantages of crown prosthesis	5 hrs lecture 1 hr tutorial
- Describe basic concept TMJ movement, natural and artificial occlusion - Examine a patient for crown prosthesis. - Enumerate the caries, periodontal status & crown-root ratio for treatment planning. - Select a tooth for full/partial veneer crown/post retained crown.	- Fundamental of occlusion - Examination, diagnosis and treatment planning	5 hrs lecture 2 hrs clinical 2 hrs practical
Describe biological, mechanical and aesthetic consideration during tooth preparation.	Principles of tooth preparation	3 hrs lecture 2 hrs clinical 2 hrs practical
- Describe the technique of impression. - Prepare of cast	Impression technique, model preparation	2 hrs lecture 2 hrs clinical 2 hrs practical
	Preparation of provisional restoration	
Perform construction of inlay, onlay, partial veneer crown and full veneer crown.	Construction of intra-coronal and extra-coronal restoration	3 hrs lecture 2 hrs practical
- Describe the luting agents - Innumerate the cementation procedure.	Cementation	1 hr lecture 1 hr tutorial 2 hrs practical
Perform the periodic check up of the patient.	Post operative care	1 hr lecture 1 hr tutorial 3 hr practical

Learning Objectives	Contents	Teaching Hours
- Define FPD. - Describe the indication, contraindication, advantages & disadvantages of fixed partial denture & Perform FPD	- Fixed partial denture: - Introduction, terminology, indication, contraindication, advantages and disadvantages of fixed partial denture prosthesis	6 hrs lecture 2 hrs tutorial 4 hrs clinical
Student should be able to- - Examine a patient for fixed partial denture prosthesis. - Enumerate the caries, periodontal status & crown-root ratio for treatment planning.	Examination, diagnosis and treatment planning	11 hrs Lecture 1 hrs tutorial
Perform wax pattern, casting, soldering, finishing and polishing of FPD.	Laboratory procedures involved in the fabrication of fixed partial denture	7 hrs lecture 2 hrs tutorial 8 hrs practical
Describe cementation procedure.	Cementation	1 hr lecture 3 hr clinical 2 hr practical
Perform the periodic check up of the patient.	Post operative care	1 hr lecture 2 hr clinical 2 hr practical

Learning Objectives and Course Contents in Prosthodontics

To be taught in 4th phase

Maxillofacial Prosthodontics

Learning Objectives	Contents	Teaching Hours
Student should be able to- ● Define maxillofacial prosthesis ● Describe the defect areas, ● Describe types of maxillofacial prosthesis, ● diagnose the cases and treatment planning, ● Innumerate the materials used in maxillofacial prosthesis.	● Introduction ● Applied Anatomy, physiology and pathology ● Speech consideration ● Materials used in maxillofacial prosthesis ● Various maxillofacial prosthesis: Intraoral and extra-oral prosthesis (obturators, splints, stent and artificial extra-oral prosthesis) ●	3 hrs lecture 3 hrs tutorial 3 hrs practical

Learning Objectives and Course Contents in Prosthodontics (Implant Dentistry) To be taught in 4th phase

Learning Objectives	Contents	Teaching Hours
Student should be able to- ● Define implant ● Describe terminology related to implant. ● Enumerate the different dental implants and their uses.	● Introduction and Scope ● Applied anatomy, physiology, pathology, pharmacology	1 hrs lecture 2 hrs tutorial 2 hrs practical
● Diagnose and Identify the cases. ● Perform treatment planning and designing. ● Describe Osseo-integration	● Examination, diagnosis and treatment planning ● Osseo-integration and occlusion scheme	2 hrs lecture 1 hr tutorial 3 hrs clinical
● Fabricate surgical stents in the model..	● Surgical stents	2 hrs lecture 1 hrs practical
● Describe prosthodontic replacement options.	● Prosthodontic rehabilitation.	2 hrs lecture